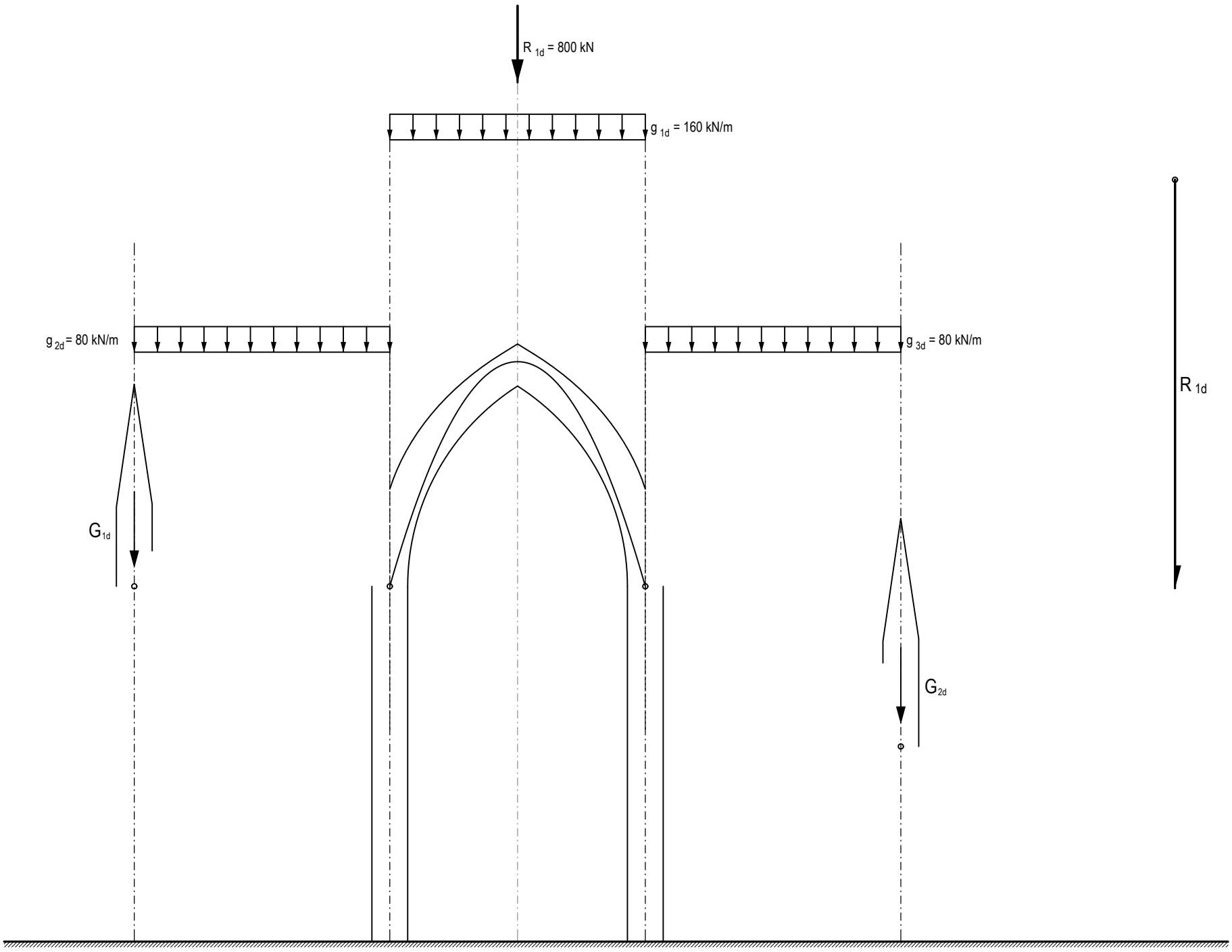
Creative Task

Cathedral

Complete the given cross section of the vault of the main nave of a cathedral with your own design of side naves and supporting walls.
Draw the inner force flow in the construction with help of the force diagram and adjust your design if necessary. The forces have to be led into the ground through the structure. Keep in mind that a stone construction works in compression only.

a) The inner force flow in the vault of the main nave is already given. First, find the geometry of the vault of the left side nave. Take into account the vertical walls inbetween the main and the side nave which define that the inner force has to be transferred vertically to the supports.
b) Then coordinate the geometry of the left supporting wall with the external force F_{1d}, which you can freely choose.
c) Apply the same procedure to design the structure of the right side nave.



form diagram 1:100

force diagram 1cm ≙ 100 kN